No	Paper Name	Year	Model	Code	Accuracy	Pros	Cons	Future
1	Biomedical Engineering and Biocybernetics Team	2020	Support Vector Machine (SVM), Linear Discriminant Analysis (LDA), and Multi-layer Perceptron (MLP)	https://github.com/biolab-put/putEMG	**Offline Accuracy:** ~90% to 95% for distinct hand gestures under ideal, static conditions	Provides high-density sEMG data (24 channels) along with a public, easy-to-use processing pipeline. Excellent baseline for multi-channel gesture classification in controlled environments	Completely recorded in a static setting, meaning it does not account for dynamic arm translations Susceptible to accuracy drops in real-world scenarios due to electrode shifts or dynamic movement	Developing adaptive deep learning models to overcome inter-session signal drift and user movement artifacts
2	Gesture Recognition via Estimation of Information Exchange between Muscles	2023	DTF MVAR Classification -Kernal SVM, KNN, CNN Subject- 2,10,30	https://github.com/AliciaFalconCaro/EMG_Gesture Recognition_DTF	Subject 2: 90.3%(Kernal SVM) in DTF Subject 10: 86.4%(KNN) in DTF Subject 30: 86.4%(KNN) In DTF	Models directional exchange between muscles	DTF connectivity not evaluated properly	DTF connectivity should be evaluated using LOSO
3	**EMG Dataset for Gesture Recognition with Arm Translation**	2025	Hierarchical Multi-label Classification baseline / Myoelectric control decoding	Examples with data read/ write, pre-pr	**Standard (same position):** ~96%	Captures 3D arm translation (3x3 grid), multi-day variations, and	Low baseline model accuracy (58%) due to multi-lab	Developing position-invariant decoding algorithms and expanding

			models	ocessing, and a simple classification pipeline are provided in detail in the following repository: https://github.com/MoveR-Digital-Health-and-Care-Hub/MoveR_AT_GREAT The data collection experiment code can be found on GitHub (https://github.com/MoveR-Digital-Health-and-Care-	**Naive Transfer (across positions):** Drops by up to 10% **Position Classification:** ~80% **Hierarchical Multi-label (HMC):** 58% **Soft HMC (gesture only):** 92%+ **Soft HMC (gesture only):** 92%	synchronous 16-channel sEMG with hand kinematics.	el complexity, with data limited only to healthy male subjects	the dataset to include female and limb-different subjects.

				Hub/posture_dataset_collection) and was implemented in Python using the Axopy package[63].				
4	**SparseEMG: Computational Design of Sparse EMG Layouts for Sensing Gestures**	2025	4 electrode-selection methods: Mutual Information, Permutation Importance, RMS-based Importance, and SHAP **Classifirs:** KNN,LR,MLP,NB,RF,SVC,XGB		94.4% with 76 electrode (RF) in PI	Demonstrates major electrode reduction	Electrode optimization is not integrated with deep graph learning.	Combine sparse electrode selection with subject-independent GNN
5	**Hand Gestures Recognition Using Convolutional Neural Networks**	2022	putEMG-force Classifier- ANN, KNN, CNN		>=95 CNN	Applies STFT to represent nonstationary sEMG signals.	Detailed structure is missing	Report CNN structure with details

Reference

1. https://biolab.put.poznan.pl

2.https://2023.ic-dsp.org/wp-content/uploads/2023/05/DSP2023-52.pdf

3.https://www.nature.com/articles/s41597-024-04296-8#code-availability

4.https://dl.acm.org/doi/10.1145/3746059.3747614

5.https://ijaem.net/issue_dcp/Hand%20Gestures%20Recognition%20Using%20Convolutional%20Neural%20Networks.pdf